



Review article

Use of information from monitoring and decision support systems in pig production: Collection, applications and expected benefits

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ABSTRACT

This paper reviews the use of information from animal-based monitoring systems, management information systems (MIS) and decision support systems (DSS) in pig production. As technology evolves, more and new types of data can be automatically collected. This increases the pace at which new systems are being developed. The collected data are processed into information, which should be used at the herd level. As these systems are being developed, there is a need to evaluate the value of the information provided to the user. In this review, the concepts of data and information are first described as well as the factors influencing the value of information, e.g. measuring devices and data/information processing. Then, after presenting the technologies and recordings for obtaining data used in the studies included in this paper, the value of information of the systems is reviewed according to their expected benefits in terms of production and management, animal health and welfare, and economical impact. The aim and value of DSS and other MIS are thereafter reviewed. Few studies have evaluated the economical benefits of using technologies to obtain new information, in terms of the number of piglets or the economical value. To quantify the value of information, it is suggested to develop a model frame with a detailed level of information, which allows one to estimate the value of including or excluding one piece of information. A more precise quantification of the value, and benefits, of information can in the future assist the development of more focused systems, in which implementation may be facilitated by reducing the uncertainty of the payback period.

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1. Introduction

During the past decades, new monitoring systems, management information systems (MIS) and decision support systems (DSS) have been developed for pig production. This development was initiated in the late 1950s, where descriptive and mathematical modeling was applied to solving agricultural problems. Numerous possibilities of application were early foreseen and discussed (Barrett et al., 1992; Frost et al., 1997). The advent of sensor technology facilitated and increased the amount of data collected. This increased the pace at which modeling and monitoring systems were developed and per se contributed to the emerging field of Precision Livestock Farming (Wathes et al., 2008).

Currently, for pig production, the information used as a basis for decision making is a combination of observations of the animals and their environment, as well as production results, which are typically monthly or quarterly reported.

The quantity and accuracy of information can contribute to have more objective measurements that can assist decision making, and as compared to visual assessment, the use of sensor technology can provide more objective and reliable data. In this context, integrated monitoring systems can combine the strengths of the stockman and the computer-based system (Frost et al., 1997). According to the authors, an integrated monitoring system approach is required, in which sensor data are integrated with information from databases, mathematical models and knowledge bases. In order to be viable, van der Stuyft et al. (1991) claimed that the developed systems should have a significant effect on production or welfare. In an early review of electronic identification of farm animals, Geers (1994) provided some aspects of the required developments and expected benefits of these systems. The author depicted examples of economical benefits associated with the use of sensors.

These systems should also take into consideration some pitfalls already reported at an early stage, such as knowing the relative importance of a problem, the development costs, or the uncertainty payback period for capital investment (Barrett et al., 1992; Wathes et al., 2008). Hence, it can be assumed that a better knowledge of the utility value associated to the information provided by new monitoring and decision support systems may facilitate the development, and potential for implementation, of more focused and efficient technologies and systems.

There is, therefore, a need to assess the value of implementing a new technology, monitoring or decision support system, or in other words, the value of the information provided by these systems. The elements that should be taken into consideration are the following: (1) what type of information (knowledge about individual or

group of animals, and/or environment) this technology can contribute with and (2) the value of this information, which depends on its potential in improving manager's decisions in the daily herd management. Whether the data or information is already available through daily registrations, or new data need to be collected and further processed as information, this process involves a cost that should be balanced with the value of information.

This paper reviews and discusses the value of information provided by monitoring, information and decision support systems for pig production. It begins with defining the concepts of data, information and value of information. This is followed by an overview of the technologies used to obtain data. Thereafter, the value of information of monitoring some given traits using animal-based monitoring systems or manually registered observations is depicted according to three types of benefits: (1) production, management and general decision making, (2) health and welfare and (3) economical value, including labor cost savings. The considered systems are animal based, i.e. monitoring climate or environment is out of this review's scope. Then, the value of information of decision support systems (DSS) and other management information systems (MIS) combining several traits is examined. Finally, the last section discusses the findings and suggests some perspectives.

2. The concepts of data, information, decision and value of information

The combination of increased herd size and the use of new technologies imply that large amounts of data can be automatically collected, and numerous registrations are made on a daily basis. In order to be used as a basis for decision making, data need to be processed into information. Fig. 1 illustrates the path from measurement to decision (left side), as previously depicted in its simplest form in Kristensen et al. (2010a). The first processing (I) converts the data into information, e.g. the physical or physiological state of an individual or a group, such as weight or illness. The second processing (II) treats the information so that it can be used for decision making. It is, for instance, the application of tools for replacement or delivery policies. This schematic overview also emphasizes that a given information can be provided by different data sources, which can themselves be collected using a variety of measuring devices (right side). Furthermore, decision support systems (DSS) or management information system (MIS) can use more or less information sources to assist in the decision process.

The value of information is influenced at several levels. In the simulation model described by Jørgensen and

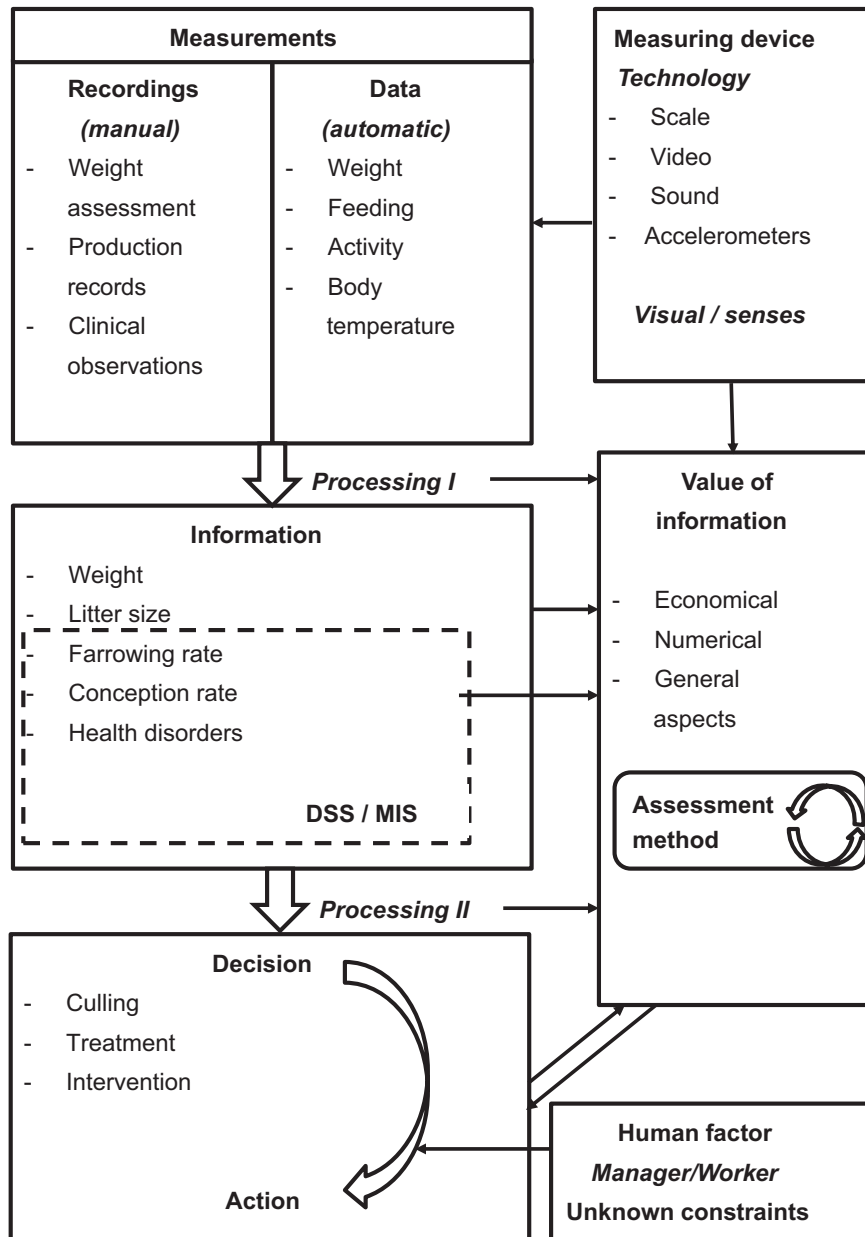


Fig. 1. The value of information set in relation to the path from on-farm measurements to decision and action. The black arrows indicate the influence of the different elements upon the value of information.

Kristensen (1995), the authors include the concept of a measuring device, which covers both the actual (referred here to as technology) and virtual devices (referred here to as visual or senses). Any measuring device, e.g. visual assessment or manual scale for assessing a pig weight, is associated with a given precision, which can affect the information, and its value. The value of information is also influenced by the type of information, and how much information are taken into account in a DSS/MIS. The path from decision to action is also influenced by a human factor. Here, risk aversion can influence the value of and decision to use (or not) the information (Byerlee and Anderson, 1982; Nadiminti et al., 1996) and the transfer

of information can be influenced by the type of organization, i.e. hierarchical or network (Sillince, 1995). Jørgensen and Kristensen (1995) also included an additional step from a suggested decision from the DSS to the actual decision by the manager, i.e. constraints unknown to the DSS and a possible irrational component. Finally, the influence of the methodological and statistical aspects should also be pointed out. Whether it is during the first processing (from measurement to information), the second processing (from information to decision) or during the assessment of the value of information, the applied methods can affect the estimated or expected value of information.

To quantify the financial benefits of electronic identification and monitoring system within animal husbandry, Geers (1994) mentions two approaches (from Dijkhuizen, 1983): partial budgeting and cost–benefit analysis. Here, the zootechnical benefits that result from the introduction of electronic identification and monitoring, and the profitability of these systems over a sufficient period of time should be taken into account. For swine production, Jørgensen (1992) suggests treating the information as a production factor, in line with, for instance, feed. In that case, the value of information should be measured on its output, i.e. the improvement in decisions. In the context of decision-making theory, Huirne (2002) points out that evaluating the results of a decision yields to new information, from which adjustments of the decision can be made. The author mentions that the role of information is most often to reduce risk and uncertainty and that the benefits of information are effectuated through better and more rapidly taken decisions. Kristensen et al. (2010b) suggest to define the value of a registration (and then information) from the actual action chosen. They assume that (i) a registration has only value if it leads to an action and (ii) the value of a registration is that it reduces the uncertainty. The value of a registration is then defined as *the expected*

utility based on actions when the registrations are made minus the expected utility when the registration is not made (Kristensen et al., 2010b). This value can be expressed as numeric weights or in economical terms. The information obtained from either monitoring systems or DSS can also be used to assist decision making, and even though not directly evaluated, can have a financial impact. Therefore, more general aspects, such as management or health will also be reported while assessing the value of information.

3. Technologies and recordings for obtaining data

The left panels of Tables 1–3 indicate the different types of data collected in the articles included in this review. The reported monitoring systems generally focused on collecting one type of data at a time.

Identification are the first stage of the development of systems monitoring systems. The technology generally used on-farm and in the reported studies is an ear-tag transponder (passive sensor), where a power antenna located, for instance at the feeding station, is used

Table 1

Value of information for monitoring systems, according to production, management and general decision making.

Identification	Facilitate development of semi-extensive systems (Geers, 1994)
Weight and improving weighing precision	Predict growth and delivery (Kristensen et al., 2012; Parsons et al., 2007; van der Stuyft et al., 1991) Control of factors, e.g. space, feed (Brandl and Jørgensen, 1996; van der Stuyft et al., 1991) Precision of weight at slaughter (Brandl and Jørgensen, 1996; Wang et al., 2008) Reduce environmental pollution (Parsons et al., 2007) Combine with information from feeding equipment (Jørgensen, 1993) Production monitoring, commercial practice (Brandl and Jørgensen, 1996; Ramaekers et al., 1995) Improve procurement plan and better match with industry (Apichottanakul et al., 2012) Indication about pig value, e.g. breeding value, lean content, yield (Wu et al., 2004)
Drinking behavior	Regulate indoor climate and improve productivity (Bird and Crabtree, 2000) Combine with other behavior analysis, as feed intake (Bird and Crabtree, 2000; Kashiha et al., 2013) Guide farmer on where to focus (Madsen and Kristensen, 2005; Madsen et al., 2005)
Feeding behavior	Detecting oestrus, combine with other information, guide farmer on where to focus (Cornou et al., 2008) Measure production efficiency and improve growth performances (Schulze et al., 2003) Determine pre-slaughter fast period, improve carcass quality (Warris and Brown, 1983)
Body temperature	Monitoring oestrus (Geers et al., 1995; Henne, 1991; Junge-Wentrup and Holtz, 1984) Monitoring farrowing (Bressers et al., 1994) Environmental temperature control (Andersen et al., 2008; Geers et al., 1997; Wouters et al., 1990) Support herdman's observation routine (Wouters et al., 1990) Focus attention on individuals in needs, especially relevant in group housing systems (Bressers et al., 1994)
Activity	Monitoring oestrus (Bressers et al., 1993; Cornou and Heiskanen, 2007; Freson et al., 1998; Geers et al., 1995; Serlet, 2004) Monitoring farrowing (Cornou and Lundbye-Christensen, 2012; Cornou et al., 2011; Erez and Hartsock, 1990; Mainau et al., 2009; Oliviero et al., 2008) Assist in daily decision making (Geers et al., 1995) and 'management by exception' (Bressers et al., 1993; Cornou and Heiskanen, 2007; Cornou and Lundbye-Christensen, 2012)
Visits to the boar	Detecting oestrus (all cited authors for monitoring of visits to the boar). Combine information with other information sources (Ostersen et al., 2010) Assist management of sows and gilts in outside pen (Blair et al., 1994) Develop systems adapted to group housing (Marchal et al., 1996; Ostersen et al., 2010)
Pregnancy	Outcome of pregnancy test can allow intensifying heat detection (Jørgensen, 1992) Outcome of pregnancy test risks reducing total number of farrowing if use for culling (Jørgensen, 1992)
Litter size/farrowing rate	Combine with other information, use in culling strategy and as production control (Bono et al., 2012, 2013; Jørgensen, 1992)

Table 2

Value of information for monitoring systems according to the value for animal health and welfare.

Identification	Prevent disease outbreaks (Geers, 1994) Reduce pneumonia incidence from 10% to 7% (Huiskes, 1991) Measuring welfare during housing and transport (Geers, 1994)
Weight and improving weighing	Detection of diseases (Brandl and Jørgensen, 1996; Jørgensen, 1993; Parsons et al., 2007; van der Stuyft et al., 1991; Wu et al., 2004) Focus on health and welfare (Parsons et al., 2007) Reduce pig stress (Wang et al., 2008; Wu et al., 2004)
Drinking behavior	Detection of disease outbreak (Kashiha et al., 2013; Madsen and Kristensen, 2005; Madsen et al., 2005)
Feeding behavior	Detecting health disorders of group housed sows (Cornou et al., 2008) Behavioral studies in different housing and feeding systems (Bornett et al., 2000; Morrison et al., 2003; Nielsen et al., 1996) Indicator of social constraint (Nielsen, 1999)
Body temperature	Improve health (Bressers et al., 1994) Reduce stress (Wouters et al., 1990) Reduce hygiene and welfare problems (Andersen et al., 2008)
Activity	Reduce perinatal mortality when continuous surveillance of farrowing (Geers, 1994) Detect movement that can lead to piglet crushing (Mainau et al., 2009) Improve health and welfare (Oliviero et al., 2008) No disturbance of individual sows for farrowing check (Erez and Hartsock, 1990)
Visits to the boar	Reduce pig stress (Buré and Houwers, 1989).
Cough	Earlier detection of disease, prevention of disease spreading and improvement of animal welfare (Exadaktylos et al., 2008; Ferrari et al., 2008) Improve diagnosis of enzootic pneumonia (Nathues et al., 2012)
Clinical signs	Risk underestimating occurrence of diarrhea (Pedersen et al., 2011)

Table 3

Value of information for monitoring systems according to the economical value, including labor cost savings.

Identification	+0.5 weaned piglets/litter and reduce non-productive days (Dopporto et al., 1990) Reduce labor time and cost when addition of monitoring (Petersen et al., 1989) Improve working conditions (Geers, 1994) +2.5 EUR/sow (Petersen et al., 1989) Reduce yearly losses per epidemic from 155 EUR (identification only) to 38 EUR (incl. monitoring) (Saatkamp et al., 1997) ^a
Weight and improving weighing precision	Reduce labor time and cost (Brandl and Jørgensen, 1996; Parsons et al., 2007; Wang et al., 2008; Wu et al., 2004) Reduce stress and physical burden (Brandl and Jørgensen, 1996) +2 to 7 EUR/pen space/year (Jørgensen, 1993) ^b +0.05 EUR/pen/week (Kristensen et al., 2012) ^c
Feeding behavior	+1.35 to 1.88 EUR/pig space using multiphase feeding (Niemi et al., 2010)
Activity	Reduce labor time and unnecessary oestrus checks (Bressers et al., 1993, 1991; Freson et al., 1998). Reduce the frequency of visits in the farrowing house for farrowing detection (Cornou and Lundbye-Christensen, 2012; Erez and Hartsock, 1990) Reduce time and labor associated with behavioral studies (Mainau et al., 2009) +200 EUR/sow/year when continuous surveillance of farrowing (Geers, 1994)
Visits to the boar	Reduce number of non-productive days (Ostersen et al., 2010) +8% improvement in oestrus detection. For a 2400 sows herds: reduce space requirement by 19 sows (7347 EUR over the life og gestation facility), reduce feeding cost (2742 EUR/year), increase income from extra pigs (54847 EUR/year) (Korthals, 1999) ^d
Cough	Substitute the use of PCR for diagnosis: 15–25 EUR per sample (Nathues et al., 2012)
Clinical signs	Highest reduction in profit margin for bone fractures in finisher pigs (55 EUR), lowest for lesion to the volar area of the feet (0.8 EUR) (Jensen et al., 2012) ^b Reduction in net return of 4 EUR/sow/year by observing 'unwillingness to stand' in pregnant sows, as compared to observing all clinical signs (Rodriguez et al., 2011) ^b
Litter size/farrowing rate	+0.1 pigs/litter using sow or parity specific information in culling strategy (Jørgensen, 1992)

^a From Bfr (Belgian franc) to EUR (euro).^b From DKK (Danish kroner) to EUR.^c From NLG (Dutch guilder) to EUR.^d From USD (US Dollar) to EUR. <http://www.oanda.com/currency/converter/>, February 2nd 2013.

to recognize the animal when close enough to the antenna. Identification systems were reviewed by Geers (1994) and further discussed by Frost et al. (1997). Most studies presenting sensor-based monitoring of the individual animal use identification systems in order to record animal-specific data.

Weight measurements can be performed manually, using a scale, or assessed visually by the farmer. To automatically weigh individual (or groups of) pigs, the technologies from the studies included in this review are the following: forelegs weighing systems (Ramaekers et al., 1995), image analysis (Brandl and Jørgensen, 1996; Parsons et al., 2007), walk trough using machine vision (Wang et al., 2008), or photogrammetry to determine the three dimensional shape of the live pigs (Wu et al., 2004). These systems allow one to measure weight within 3% (Wang et al., 2008) to 5% (Brandl and Jørgensen, 1996; Ramaekers et al., 1995) of the manually measured weights, within 2 kg of the target weight (Parsons et al., 2007) or ± 6 mm for the technology used by Wu et al. (2004). Alternatively, modeling using artificial neural network, based on, among other things, previous recordings on weight at entrance and daily gain, has been used to estimate the pig size and distribution of pigs in a herd (Apichottanakul et al., 2012).

Body temperature can be measured on different locations: ear-based temperature was reported in Andersen et al. (2008), Vermeer (1999), Soede et al. (1997), Michael (1976), Bressers et al. (1994), and as non-automated methods, the measure of vaginal or rectal temperature was reported in Junge-Wentrup and Holtz (1984) and Henne (1991). Alternatively, the thermoregulatory behavior of pigs, using image analysis, can be used as an indicator of body temperature (Wouters et al., 1990).

Drinking behavior is monitored using water flow-meters (Madsen and Kristensen, 2005; Madsen et al., 2005) or using vision technology to count the number of visits at the drink nipple (Kashiha et al., 2013).

Feeding behavior is monitored using an ear transponder that identify the individual pig at each visit to the feeder (Bornett et al., 2000) or using video-taped data (Morrison et al., 2003; Nielsen et al., 1996). At an earlier stage, Slader and Gregory (1988) used a combined feeder/weigher that record the time of feed, amount consumed and live weight of individual pigs housed in groups. Data from Electronic Sow Feeders, commonly

used on-farm for group housed sows, have been used in Cornou et al. (2008).

Pig cough are recorded using microphones (Exadaktylos et al., 2008; Ferrari et al., 2008) or by counting coughing bouts (Nathues et al., 2012).

Clinical signs for both sows (Jensen et al., 2012; Rodriguez et al., 2011) and piglets (Pedersen et al., 2011) are data manually collected by observing the individual animals.

Activity data can be automatically collected using different technologies. For sows housed in crates, infrared sensors/photocells have been used by Serlet (2004), Freson et al. (1998), Erez and Hartsock (1990), Mainau et al. (2009), Oliviero et al. (2008) and force sensors by Oliviero et al. (2008). Both for loose and crate-housed sows, accelerometers have been used (Bressers et al., 1993; Cornou and Heiskanen, 2007; Cornou and Lundbye-Christensen, 2012; Cornou et al., 2011; Geers et al., 1995; Serlet, 2004).

Visits to the boar are monitored using a ticket window (Blair et al., 1994; Buré and Houwers, 1989; Korthals, 1999; Ostersen et al., 2010) or a separate detection area (Bressers et al., 1991; Marchal et al., 1996).

Pregnancy data (Jørgensen, 1992) as well as *litter size/farrowing rate* data, as used in Bono et al. (2012, 2013), Huirne et al. (1988, 1991b, 1993, Kristensen and Søllested (2004a, 2004b), are obtained from manual registrations made on regular basis by the farmer.

Information obtained from observing clinical signs and measuring body temperature a non-automatic way is included in this study. Even though this information is nowadays manually collected, there is a potential for automatic collection in developing new systems. Also, data about pregnancy, farrowing rate, and litter size are included since some of this information is obtained through data processing and can be in a second step used within some DSS. The concept of MIS includes the monitoring systems (here, animal-based) using technologies, the processing of a single type of data (for instance litter size) into information and also processing of several types of data. Whereas monitoring systems and MIS based on a single data source are reviewed in the next three sections, the latter type of MIS will be included together with the DSS reported in Section 7.

4. Value of information for production, management and general decision making

Table 1 presents the value of information which can be associated with production, management and general decision making, according to the obtained data or

parameters described in [Section 3](#). Benefits common to several monitoring systems or MIS using a single data type are first presented (and set in *italics*), then benefits specific to each method are introduced.

As common use and benefit in terms of management, *oestrus detection* can be performed by monitoring body temperature (ear base, rectal and vaginal), activity, visits to the boar and feeding behavior. For monitoring methods for oestrus detection using body temperature and visits to the boar pen, reference is made to the review by [Cornou \(2006\)](#). [Cornou and Heiskanen \(2007\)](#) developed a method based on accelerometers for detecting oestrus for group housed sows, which consisted in monitoring the deviation in the activity level prior oestrus. Finally, the method developed by [Cornou et al. \(2008\)](#) relied on monitoring individual feeding behavior, more specifically individual feeding rank of group housed sows, using data from Electronic Sow Feeder.

Body temperature and activity are also both used to develop methods for *detecting the onset of farrowing*. [Bressers et al. \(1994\)](#) monitored ear base temperature and reported a rise in sow's temperature between 6 and 12 h before farrowing. Numerous studies focused on detecting changes in activity pattern, reflecting nest building behavior, prior parturition. [Erez and Hartsock \(1990\)](#) used an infrared photocell system for sows in crates prior to parturition. The use of photocell to measure sow's activity around farrowing is reported in [Mainau et al. \(2009\)](#) and [Oliviero et al. \(2008\)](#), the latter also using a force sensor to measure the overall movement of the sows. The study carried out by [Mainau et al. \(2009\)](#) focused on the sow's position changes around farrowing using Standing Lying Sensors, i.e. photocells placed near the forelegs of each sow. [Oliviero et al. \(2008\)](#) report a higher activity ($p < 0.05$) in the 24 h prior farrowing. They also suggest that monitoring activity, which is the result of nest building behavior, is more reliable than monitoring other physiological parameters, since this behavior is specific to the approach of farrowing. [Cornou et al. \(2011\)](#) and [Cornou and Lundbye-Christensen \(2012\)](#) use acceleration data and suggest two methods to monitor deviations in the daily (modeled) activity patterns of individual sows. They also point out that monitoring individual activity using accelerometers is also adapted to group housing, as compared to other methods (infrared sensors or photocells), more adapted to individual housing in crates.

Benefits in terms of *control of indoor climate* are mentioned for body temperature and drinking behavior. For body temperature, [Wouters et al. \(1990\)](#) developed a software using image analysis for environmental temperature control by quantifying the thermoregulatory behavior of pigs. According to the position (relative to the sleeping area) and behavior of the piglets, it can be suggested to adjust the temperature. According to [Geers et al. \(1997\)](#), the temperature of individual animals can also supply information for air temperature. [Andersen et al. \(2008\)](#) used ear skin temperature (EST) to measure the thermal state of pigs. Their results indicated that the general activity of the pigs influences both the EST and lying behavior. Therefore, if these should be used as an index of the thermal comfort, the general activity level of the

pigs should be taken into consideration. [Bird and Crabtree \(2000\)](#) discuss the potential of monitoring the production performance in real-time, and report results based on empirical data obtained from production systems in the UK and the USA. The authors point out the relationship between water use and air quality, and suggest therefore to use water consumption to regulate the indoor climate.

Control of feed intake and production efficiency are mentioned as common benefits of information about weighing, feeding behavior and body temperature. The use of feed intake and corresponding weight gain for measuring the production performances is straightforward. Feeding behavior is tightly related to production performances and information about individual feed intake has a clear value for monitoring and improving growth performances ([Schulze et al., 2003](#)). [Kristensen et al. \(2012\)](#) mention that weighing allows one to update the estimated growth potential, local deviations from the growth curve and the homogeneity of the pigs in order to predict their future growth and delivery. According to [van der Stuyft et al. \(1991\)](#), information about growth rate allows better planning of diet composition, weight of feed supplied to each pig, prediction of marketing date, and control the conformation of the pig. [Parsons et al. \(2007\)](#) described a real-time system for the integrated control of pig growth, where pig weight is estimated by visual analysis and feed intake is recorded daily. Their results suggest good potential for estimation of the growth rate. An additional benefit of this system, as reported by the authors, is reduced environment pollution by adjusting feeding plan according to the weight. [Brandl and Jørgensen \(1996\)](#) performed weigh estimation using image analysis and report that weighing is important with regard to the control of factors which affect the output of the herd, e.g. space allowance and feed ration (when restricted feeding is used) and, as also mentioned by [Wang et al. \(2008\)](#), the precision of weight at slaughter.

The information obtained from monitoring a given production trait (weight, drinking behavior, visits to the boar) can also be used in *combination with the information from another trait* and as such increase the value of the obtained information. For instance, [Jørgensen \(1993\)](#) suggests that weight information can be combined with information from feeding equipment. [Kashiha et al. \(2013\)](#) use vision technology to automatically monitor drink nipples visits for growing pigs. They use a data-based dynamic model to quantify water use and compare it to half-hourly duration of visits at the drink nipples, monitored by image analysis. The authors suggest that vision technology can be used to monitor drinking behavior with other behavior, e.g. feed intake, and is as such more advantageous as compared with conventional water meters. In this example, the technology provides different types of information and as such reduce the cost of information. [Bird and Crabtree \(2000\)](#) report that monitoring water use, also combined with feed intake, and water patterns can improve productivity. [Cornou et al. \(2008\)](#) suggest to further develop their monitoring system for detecting oestrus, lameness and health disorders by including additional information in their model, such as daily feed intake, body activity and results from pregnancy

test. [Ostersen et al. \(2010\)](#) also suggest that additional information should be included to the visits to the boar in order to improve the specificity of the oestrus detection. Examples of additional information reported by the authors are pregnancy tests, farrowing rate and proportion of sows entering oestrus at day 21. In that case, the additional information could reduce the number of false positives and ultimately the labor time associated with checking sows for oestrus. [Bono et al. \(2013\)](#) developed a model and monitoring system for farrowing rate. They suggest to combine this information with information about litter size ([Bono et al., 2012](#)) and mortality rate in order to assist the farmer in management decision making and reduce economic losses.

Another common benefit is to support *daily decision making and a tool to guide the farmer on where to focus*. This is mentioned in several studies reporting results of monitoring drinking/feeding behavior ([Cornou et al., 2008](#); [Madsen and Kristensen, 2005](#); [Madsen et al., 2005](#)), body temperature ([Bressers et al., 1994](#); [Wouters et al., 1990](#)), and activity ([Bressers et al., 1993](#); [Cornou and Heiskanen, 2007](#); [Cornou and Lundbye-Christensen, 2012](#); [Geers et al., 1995](#)).

Facilitation of group-housing, development of semi-extensive and outside systems are reported as benefits of identification systems ([Geers, 1994](#)) and monitoring of the visits to the boar ([Blair et al., 1994](#); [Ostersen et al., 2010](#); [Marchal et al., 1996](#)).

Other information value specific to weighing (and feeding) data is to improve the procurement plan to better match the industry demands ([Apichottanakul et al., 2012](#)). For this study, a model allows one to predict the average weight and weekly size distribution of finishing pigs using inputs such as average weight and age of piglets, average feed intake and number of fattening days. The drawback is that weight estimation is not in real time. According to [Wu et al. \(2004\)](#) the advantage of measuring pigs in 3D is that the shape, or confirmation, of a live pig gives indication about its value (e.g. breeding value, lean content, shape and yield of different cuts of meat) and its nutritional requirements. Finally, [Ramaekers et al. \(1995\)](#) point out that the knowledge of pig's body weight is important not only in commercial practice but also in research, e.g. where measurements should be undisturbed (e.g. for climate respiration chambers).

Determination of the pre-slaughter fast period and improvement of the carcass quality are mentioned by [Warris and Brown \(1983\)](#) (in [Slader and Gregory, 1988](#)) as expected benefits of monitoring feeding behavior, more particularly the normal periods of maximum feed intake.

For regular on-farm registrations or measurements, [Jørgensen \(1992\)](#) mentioned that the value of the information from a pregnancy test depends on the use of the information: decision about culling or focus on heat detection for sows to re-inseminate, or used in production control. It can have a high positive value if it is used to intensify the heat detection for sows detected as non-pregnant. On the other hand, using the outcome of a pregnancy test to cull sows risk reducing the total number of farrowing in the herd, lower the average age with a corresponding lower average litter size and as such reduce

the farmer's income. For litter size and farrowing rate registrations, the mentioned benefits are the use in replacement decision and as a production control.

5. Value of information for health and welfare

[Table 2](#) presents the benefits of monitoring specific traits or data in terms of health and welfare. As in the previous section, benefits common to several monitoring systems or MIS are first presented, followed by benefits specific to each method.

Prevention and (earlier) detection of health disorders are expected benefits of several monitoring systems, i.e. for identification, monitoring of body temperature, weighing, drinking, feeding and cough. According to [Geers \(1994\)](#), a better knowledge of relative risks, through electronic identification and monitoring, may prevent disease outbreaks within eradication programmes for husbandry systems. [Bressers et al. \(1994\)](#) mentioned that monitoring body temperature may improve health. According to several authors, automatic weighing can improve disease surveillance ([Brandl and Jørgensen, 1996](#); [Jørgensen, 1993](#); [Parsons et al., 2007](#); [van der Stuyft et al., 1991](#); [Wu et al., 2004](#)). This can be obtained by monitoring if the growth rate is lower than expected ([Jørgensen, 1993](#)), by providing more time to focus on health and welfare ([Parsons et al., 2007](#)), or by monitoring leg weakness using the measure of pig in 3D ([Wu et al., 2004](#)).

For growing pigs, drinking behavior has been used to monitor intestinal disorders ([Kashiha et al., 2013](#); [Madsen and Kristensen, 2005](#); [Madsen et al., 2005](#)). [Madsen et al. \(2005\)](#) and [Madsen and Kristensen \(2005\)](#) report that monitoring drinking behavior allows an earlier detection of disease outbreak and as such avoids losses in terms of reduced growth rate and increased mortality. The authors suggest moreover that this monitoring should be included in a larger system for farm management and is useful in general research on animal behavior. Also [Kashiha et al. \(2013\)](#) suggest that automatic monitoring of drinking behavior can help to improve animal welfare, by earlier detection of abnormal behaviors. The monitoring system described by [Cornou et al. \(2008\)](#) uses feeding behavior for detecting lameness and health disorders for group housed sows.

Pig cough can be a sign of respiratory disorder or infection of the respiratory system, and systems have been developed to automatically monitor health of growing pigs ([Exadaktylos et al., 2008](#); [Ferrari et al., 2008](#)). The reported benefits of this monitoring tool are an early warning of a sick cough hazard and a 24-h monitoring of pig health. This may prevent disease spreading (by earlier treatment) and as such improve animal welfare. Results reported by [Nathues et al. \(2012\)](#) support the interest in monitoring pig coughs. In their study, results indicated that quantitative assessment of the onset of coughing (using a coughing index) improves the diagnosis of enzootic pneumonia. A reduction and diagnostic improvement of pneumonia is also mentioned for identification systems. According to [Huiskes \(1991\)](#) (in [Geers, 1994](#)), information about the lung–liver system returned to the farmer after each batch (instead of after every four months) would reduce pneumonia incidence from 10% to 5%.

Monitoring stress and welfare are reported as associated benefits of several monitoring systems. Electronic identification and monitoring may facilitate the measurement of animal welfare during housing and transport (Geers, 1994). An additional benefit of automatic weighing is according to Wang et al. (2008) and Wu et al. (2004) to reduce pig stress associated to manual weighing. Monitoring body temperature may allow reducing potential stress due to non-optimal temperature (Wouters et al., 1990) and reduce hygiene and welfare problems (Andersen et al., 2008). Buré and Houwers (1989) study the sows' visits to the boar in the context of developing automation systems for group housed sows. The authors report some general advantage of automation systems for group housing as being: a better control of sow housing, a decrease of chronic stress and a lowering of the aggression level.

Studies monitoring activity in the farrowing house report some benefits of the monitoring systems, as *reducing perinatal mortality* (Geers, 1994) and in particular piglet crushing, by detecting sow movements that can lead to piglet crushing (Mainau et al., 2009). More generally, according to Oliviero et al. (2008), detecting the onset of farrowing can improve animal health and welfare through improved surveillance around parturition while Erez and Hartsock (1990) point out that automatic monitoring of the onset of parturition may avoid to disturb individual sows at the approach of farrowing.

Monitoring feeding behavior is according to some authors particularly suitable for *behavioral studies* and can also be used as an indicator of social constraints (Nielsen, 1999). The behavioral studies have monitored individual feeding behavior (frequency, duration, consumption per visit) of growing pigs for comparing, for instance, the effect of group size and housing systems (Bornett et al., 2000; Morrison et al., 2003) or pig:through ratio (Nielsen et al., 1996).

Finally, a particular example is the monitoring of clinical signs which can be associated with a negative value of information. For growing pigs, a recent study indicates that, due to the large number of false negatives, observing clinical signs for identification of diarrhoea for individual pigs is not recommended, as it will underestimate the occurrence of diarrhoea (Pedersen et al., 2011), and therefore has a detrimental effect on animal health.

6. Value of information in terms of economical value, including labor cost savings

Few studies have reported the economic value of information of monitoring specific traits or parameters. These studies are reported in Table 3, together with the ones mentioning a reduction of labor costs as a consequence of a reduction of labor time.

As a general matter contributing to a positive economical value of information, identification systems (Doporto et al., 1990) and monitoring visits to the boar to detect oestrus (Ostersen et al., 2010) may *reduce the number of non-productive days*. Another important aspect of automated systems is a *reduction of labor*. According to Petersen et al. (1989), identification systems associated with monitoring

systems (especially physiological monitoring) can reduce labor time and cost. Improvement of working conditions, by not having to restrain the pigs for identification, is mentioned by Geers (1994) for identification systems. In studies monitoring pigs' weight, several authors also point out the reduction of labor time that would before be associated to manual weighing (Brandl and Jørgensen, 1996; Wu et al., 2004; Parsons et al., 2007; Wang et al., 2008). In addition, Brandl and Jørgensen (1996) emphasize that automatic weighing reduces the physical burden associated to manually weighing the pigs and estimated the time of weighing the pig as 3–5 min for heavy pigs. A reduction of labor time is also mentioned in several studies monitoring activity: a reduction of labor time and unnecessary oestrus check (Bressers et al., 1993; Freson et al., 1998); reduction of frequency of visits in the farrowing units for farrowing detection (Cornou and Lundbye-Christensen, 2012; Erez and Hartsock, 1990); or a reduction of the time and labor associated with data analysis in behavioral studies (Mainau et al., 2009).

Generally, production losses can be reduced through a better health monitoring and improvement in management, as seen in the previous sections. Nevertheless, few studies have attempted to calculate an economical value to the use of identification systems, monitoring systems and MIS. According to Petersen et al. (1989) (in Geers, 1994), applying automatic identification can lead to an information gain of 2.5 EUR per sow. As previously mentioned, they report that this gain may increase when physiological monitoring is also introduced, especially due to labor cost savings. Electronic monitoring can also help reducing the number of non-productive days and increase by 0.5 the number of weaned piglets per litter (Doporto et al., 1990 in Geers, 1994), equivalent to 1.3 extra piglets per sow per year, corresponding to an additional 125 kg fattening pigs per sow per year. Saatkamp et al. (1997) compared four national identification and recording (I&R) systems for the Belgian pig industry, and assessed their economic impact for control of classical swine fever (CSF). The simulated yearly economic losses per epidemic ranged from 155 EUR (using identification systems only) to 38 EUR (when identification systems are used with monitoring systems). Their results indicate a high positive value of the information provided by using computerized data storage and user friendly documents.

Jørgensen (1993) estimated the value of increased weighing precision when selecting pigs for delivery to the slaughter house. The author compared different weighing precisions according to the number of pigs per pen, using flexible delivery according to the pigs weight. Results are compared to a "batch delivery", where all pigs in the pen are delivered after a fixed production period, i.e. without weighing. The economic return is reported as "present value of expected return per pen space" (PV) for the whole planning period. The highest reported PV corresponds to one pig per pen and the highest weighing precision: 933.2 compared to 868.1 EUR when batch delivery, i.e. 7.5% (or 65.1 EUR) increase of PV. For pens of 16 pigs, a very precise weighing only improve the PV of 20.4 EUR, compared to batch delivery. Considering for instance an annual interest of 10%, the reported values

correspond to an annual increase of the expected value ranging from 2.04 (one pig/pen) to 6.51 EUR (16 pigs/pen). Moreover, results indicated that the value of high weighing precision decreases markedly with the increasing number of pigs per pen, and is generally low. In terms of investment in weighing equipment, the author suggests that the maximum cost for weighing each pig in each production cycle is 2 EUR, corresponding to 206 EUR for the cost of a weighing equipment per pen. In accordance with Jørgensen (1993), Brandl and Jørgensen (1996) show that the random influence of feeding and drinking makes it irrelevant to have a high weighing precision. The authors suggest furthermore that weighing using image analysis is promising due to the low price of the standard equipment and the fact that numerous pigs can be weighed at once.

For optimization of slaughter pig marketing, Kristensen et al. (2012) developed a decision support system where a model for on-line weight assessment serves as input to the system. They compared different scenarios for delivery: (i) flexible: new piglets are inserted when the pen is empty and (ii) constant: new piglets are inserted at regular intervals. Under different weighing precisions, they reported the net return per pen and per week. Results showed that by improving the precision of weighing measurements (reducing the standard deviation of the measurement error from an arbitrary 3 kg to an unrealistic 0 kg) only improved the net return per pen per week from 11.93 to 11.98 EUR under flexible piglet supply, and from 11.60 to 11.65 EUR under constant piglet supply. This corresponds to 0.42% and 0.43% increase in net return per pen per week.

Modeling work has also been performed to optimize feeding and hence the precision of pig weight estimation. Niemi et al. (2010) used dynamic programming model and suggest that the use of multiphase feeding, as compared to two phase feeding, may result in an increase of annual return by 1.35–1.88 EUR per pig space unit.

Continuous surveillance of farrowing, would, according to Geers (1994), result in a gain probably higher than 200 EUR per sow per year. Korthals (1999) reports that the use of electronic oestrus detection equipment improves oestrus detection by 8%. The author also reports the corresponding benefits of this improvement, for a 2400 sows herd: this improvement corresponds to 19 sow places (7347 EUR over the life of the gestation facility), a reduction of feeding cost of 2742 EUR/year and an additional income from extra piglets of 54 847 EUR/year.

Nathues et al. (2012) report that, in some cases, monitoring pig coughs can substitute the use of PCR for detecting *Mycoplasma hyopneumoniae*, and therefore reduce the cost associated to the diagnosis by PCR (15–25 EUR per sample).

When observing clinical signs, Jensen et al. (2012) quantified the impact of lameness, both on the welfare and profitability of finisher pigs. By using expert opinions, the authors quantify the impact of a total of nine different causes of lameness on pain and productivity. Results indicated that pigs with bone fractures had the highest reduction in average daily gain and the highest increase in feed conversion ratio. The highest reduction in profit margin was in pigs with bone fractures (–55 EUR),

followed by lameness caused by *E. rhusiopathiae* (–25 EUR). The lowest reduction was associated with *M. hyosynoviae* arthritis (2.8 EUR) and lesions to the volar area of the feet (0.8 EUR). Furthermore, the author reported a rather linear relationship between the impact on pain (with bone fractures being the highest, and lesion to the volar area being the lowest) and the loss in profitability. Therefore, since reducing the occurrence of bone fractures has the largest positive effect, both for welfare and productivity, the authors report that it is important to distinguish between bone fractures and the other causes of leg disorders.

For sows, Rodriguez et al. (2011) investigated the effect of observing clinical signs on the optimal replacement policy, while previous sow replacement models used directly observable variables, such as litter size. The authors estimated the economic value of observing clinical signs of individual sows by using two fictitious herds with low (LR) and high risk (HR) of involuntary culling of sows. They defined 15 different scenarios, according to the number of observed clinical signs serving as a basis for developing a weak sow index (WSI): unwillingness to stand, lameness, vulva bite, body score condition, vulva color, and whether they are observed in the gestation or lactation period. As compared to a basic scenario, where all clinical signs are observed in both periods, the reduction in net returns according to the 15 scenarios ranges from values very close to zero (to refrain from observing “unwillingness to stand” in pregnant sows, for LR and HR) up to approximately 4 EUR per sow per year (to observe only “unwillingness to stand” in pregnant sows, for HR). Moreover, they reported that observing clinical signs in the gestation period may allow for better detection of weak sows than observing them in the lactation period.

Jørgensen (1992) suggests that the registration of parity specific litter size is one of the most important registrations from a production control point of view. The information about litter size can be directly used for replacement decision: if a sow obtains a litter size lower than expected at a given parity, it should be replaced. The value of this information can then be expressed as expected outcome or as improvement on litter size. The author also reports that a culling strategy using sow specific information or only parity specific information is low, i.e. less than 0.1 pigs per litter, which corresponds to the maximum possible benefit using the culling strategy. Hence, according to the author, DSS for sow culling needs not to include the variance between sows with respect to litter size, since this information has a low value.

Bono et al. (2012) developed a model and monitoring system for litter size where the individual sows' properties are used to compute: (1) a sow value, which indicates the difference (in the number of piglets) between the individual sow result and the herd parity-specific mean and (2) an economic value, or “retention pay-off”. Results indicate that even though a sow has a positive value (numerical index), the retention pay-off (economic value) may not always be positive, due to the observed reduction of litter size after parity four. Detected deviations in the production plan can lead to changes in the future

production plan, which can also have an indirect economic impact.

7. Sources of information, types and value of DSS and MIS

Table 4 presents an overview of the aims of the DSS and other MIS for pig production included in this study (top panel), as well as studies assessing the impact of DSS (middle panel). The economical value of the information provided by the reported systems is shown in the bottom panel.

As illustrated in Fig. 1, DSS and some MIS combine several sources of information, of which some of them may be obtained from monitoring systems. The type and quantity of information included in the MIS and DSS vary according to the purpose of the system. Few examples of the information obtained from the above monitoring systems can be mentioned. A large part of the developed DSS has focused on replacement strategies and information about litter size, as discussed in the previous sections, is used in several sow replacement models (Huirne et al., 1988, 1991b, 1993; Kristensen and Søllested, 2004a, 2004b). Furthermore, Rodriguez et al. (2011) suggest to use the WSI developed by observing clinical signs since it highly influences the optimal replacement strategy. For slaughter pig marketing (Kristensen et al., 2012), information about the observed weight, automatically monitored using image analysis, is included in their model.

The aims of the studies included in this review can be categorized as the following: replacement decision, production control, strategic planning and investment, optimization of disease control and delivery policies.

7.1. DSS for sows' replacement

Dijkhuizen et al. (1989) used the PorkCHOP replacement model to evaluate the economical impact of culling strategies, and compared the model recommendations to the farmers' actual decisions. They report that the highest economic losses were found in cases of lameness or leg weakness (83 EUR per culled sow). Premature disposal was on average 56 EUR per culled sow.

Huirne et al. (1988, 1991b, 1993) describe a stochastic dynamic model designed on a personal computer, which determines the optimal replacement policy for sow herds. Besides optimizing the replacement policy, the model calculates the expected total extra profit, as economic index, when retaining a sow under optimal lifespan instead of replacing her immediately, namely the retention-pay-off (RPO). This allows a ranking of individual sows, in terms of future profitability, and can be used as a decision support for culling decisions. Their results indicate an average herd life of 5.5 parities under optimal replacement decisions. Under this optimal decision, the present value of expected net returns, calculated as an annuity, is 387 EUR per sow per year.

An integrated decision support and expert system, based on the above mentioned dynamic model, was developed and validated (Dijkhuizen et al., 1990; Huirne et al., 1991a). This system, CHESS (Computerized Herd Evaluation System for Sows), analyzes individual sow-herd performance within an economic framework. Alternative production and price levels were used in a sensitivity analysis of the DSS part of the system. Results show that the "gross returns minus feed costs per sow per year" was affected by changes in litters per sow per year, litter size and feeder pig price. Moreover, the value of an increased day of farrowing interval was 1.5 EUR and for an extra pig born alive was 23 EUR. They suggest that while sensitivity analysis consists of modifying one variable at a time, a group of variables must be varied for simulating strengths and weaknesses in practice.

Later on, Huirne and Hardaker (1998) developed a multi-attribute utility model, based on the previous work using a stochastic dynamic programming model for sow replacement decisions (Huirne et al., 1988, 1991b, 1993). This model includes six attributes, of which one of them, RPO, is obtained from their previous work. Using a sample of 10 sows for a given week, results from this study indicate that the ranking of sows obtained from the multi-attribute model differs considerably from ranking based on the RPO only. Results from the expanded model appeared in accordance with experts opinions whose

Table 4

Aims of the DSS and other MIS and economical value reported in the studies.

Aim
Replacement decision (Dijkhuizen et al., 1989, 1990; Huirne and Hardaker, 1998; Huirne et al., 1988, 1991a, 1991b, 1993; Kristensen and Søllested, 2004a, 2004b; Kirchner et al., 2004; Plà et al., 2004; Pomar and Pomar, 2005)
Production control (Jørgensen, 1985)
Strategic planning, investment (Backus et al., 1995a, 1995b)
Optimization of disease control and delivery policy for slaughter pigs (Toft et al., 2005)
Optimization of delivery policy for slaughter pigs (Kristensen et al., 2012)
Assess impact of DSS
Assess impact of DSS, compare utility values (Jørgensen and Kristensen, 1995; Versteegen and Huirne, 2001; Versteegen et al., 1995a, 1995b)
Economical value
–83 EUR for culling due to lameness or weak legs and –56 EUR for premature culling (Dijkhuizen et al., 1989) ^a
387 EUR/sow/year as present value of expected annual net returns (Huirne et al., 1991b)
+1.5 EUR for an increase day of farrowing interval and +23 EUR for an extra pig born alive (Dijkhuizen et al., 1990; Huirne et al., 1991a;
+6–7% total income using production control (Jørgensen, 1985)
+0.56 piglets/sow/year (Versteegen et al., 1995b)
+1.41 to –0.69 piglets/sow/year (Versteegen and Huirne, 2001)

^a From NLG (Dutch guilder) to EUR; <http://www.oanda.com/currency/converter/>, February 2, 2013.

preferences were not used to construct the model. The authors concluded therefore that the multi-attribute utility model can be used to determine the optimal replacement policies for sows.

Kirchner et al. (2004) applied a decision tree technique to pig farm data set in order to improve the detected differences and weak points in farm management. They applied the technique on the sow replacement decision problem, using data from two sow herds. This technique makes it possible for the farmer or the consultant to get a graphical overview of the sow performance and culling strategy and compare strategies between herds.

Kristensen and Søllested (2004a, 2004b) present a sow replacement model in which methodological improvements, such as multi-level hierarchical Markov processes with decisions on multiple time scales and efficient methods for parameters estimation, are implemented. The authors use a model describing litter size, based on Toft and Jørgensen (2002), for which parameters are estimated at the herd level. This information allows one to compare economical results of and between herds, under optimal replacement policy. When comparing retention pay-off values, results indicate that herd-specific parameters should be used in replacement models.

Plà et al. (2004) developed a sow herd DSS that provides output in terms of technical and economic indices, herd structure and the sensitivity analysis of strategic parameters. The DSS allows one to estimate the expected farm net revenues according to simulated reproduction strategies and optimal replacement guidelines. For a review of the mathematical models for sow herd management, a reference is made to Plà (2007).

Pomar and Pomar (2005) developed a knowledge-based decision support system to improve sow farm productivity (CULLSOW), through early identification of sows showing low prolificity. Among other things, the software uses information from first and second parities and qualitative reasoning techniques. It classifies sows with regard to their expected prolificity (“low”, “acceptable”) together with an indication of the certainty value (“sure”, “most likely”, “likely”). Results indicate that, as compared to a reference method, CULLSOW allows one to reduce the number of culled sows (48–64% less sows according to the certainty level used) for the same increase in herd productivity.

7.2. Other DSS and MIS

An early study by Jørgensen (1985) evaluated the value of the production control in Danish herds using 100 sow herds in the Danish field test organization (Pedersen et al., 1995). The production control consists of registrations of several traits which are presented to the farmer as quarterly averages. The underlying idea of production control is to get a better knowledge of the expected future level, by observing the production traits during a period. Thereafter, the deviation of a given trait from its expected level can be weighed with its marginal value and an estimate of the economic value of the deviation can be obtained. According to Jørgensen (1985), if a decision for change in the production plan is made, then the corresponding

improvement in income corresponds to the value of the production control. Results indicated that the registrations in the Danish efficiency control could be expressed from 16 major key figures. This study showed an overall improvement of 6–7% in total income using the production control. Seven out of sixteen key figures accounted for almost all the values of the control, and were, starting from the highest marginal improvement: piglet mortality, usage of farrowing department, litter size in parity 1, 2, 3 and higher, and pregnancy rate. An important aspect discussed by Jørgensen (1985) was that improvement could be obtained by fewer registrations than currently used, and that, instead of performing a detailed control of individual performances, emphasis should be on controlling the whole herd's production.

Backus et al. (1995a) assessed the impact of a DSS on recommendations given by extension officers on pig farmers. The Investment Simulation Model (ISM) described in Backus et al. (1995b) is a stochastic model using Monte Carlo simulation, which uses data from individual farms and average data representing Dutch pig farms. The study included three farms, for which three sets of recommendations were prepared: (1) without using ISM and based on a farm visit to collect data, (2) with ISM, based on a farm visit to collect data and (3) with the use of ISM, and without visit to the farm. The differences in the recommendations made from the three scenarios laid mainly in the advice about farm extension, which was advised for all three farms using the ISM whereas considered as not feasible without the use of ISM. An explanatory factor is that the ISM had a longer time horizon. Aspects as replacement strategies, manure costs and family expenditures were largely ignored without ISM, as compared with ISM. Also, difference in time required for preparing the advice was reduced while using ISM (80% of the time required without).

For slaughter pigs, Toft et al. (2005) developed a framework for decision support related to infectious diseases. The authors illustrate the model using the example of Swine Influenza where threshold limits were defined over time and according to the number of sick pigs. This model assists, among other things, in the decision for vaccination, for different medication costs, and in optimization of delivery policy, as the model of Kristensen et al. (2012) mentioned in the previous sections.

Few studies have compared the utility value of DSS and MIS, or evaluated the impact of applying these systems. For instance, the simulation model developed by Jørgensen and Kristensen (1995) can be used for comparing the utility value of different measuring devices, Belief Management Systems (BMS) and DSS. The authors suggest that such a model can help providing sound advices to farmers using new equipments and can be used as a basis for testing DSS in different areas of pig production.

Verstegen et al. (1995a) review the evaluation approaches for estimating the economic value of MIS in agriculture. The authors indicate that the cost–benefit analysis, used for evaluate investment calculation, is difficult to apply to MIS evaluation, due to the nature of these systems, which affects a large range of activities and which benefits depend a lot on the users. They identify two approaches: normative

and positive approaches. While normative approaches have limited potential, the authors suggest that positive approaches, which analyze the changes in production under the influence of MIS use, depend on the availability and quality of field data, and the type of research design. Another aspect is that MIS focus mainly on a decision support function, and not on processing and operational functions. Hence, the benefits of MIS are mainly reflected by improvements in decision making, which are not straightforward to identify.

In [Verstegen et al. \(1995b\)](#), the authors quantify the economical impact of using MIS with panel data. The authors combined data from two survey studies. They used mixed effect models, where the effects of introducing MIS were analyzed both within and over farms, controlling for self-selection bias and changes over time. They report that farms adopting MIS increased their production by 0.56 piglets per sow per year. This corresponds to a return to investment between 220% and 348%, and implies, for a 165-sows farm, an increase in profits between 7.7% and 8.7%.

[Verstegen and Huirne \(2001\)](#) study the impact of farm management on the value of MIS, by comparing farm developments before and after the use of MIS. They reported a positive correlation between the management levels of sow farmers and the MIS value ($r=0.35$; $p=0.02$). The effect of using the MIS, as an average number of piglets (raised to about 25 kg) per sow and per year, is presented for four styles of farming. The effect was 1.41 for the 'Entrepreneur', 0.42 for the 'Manager', 0.49 for the 'Pig farmer' and -0.69 piglets per sow per year for the 'withdrawer' style of farming. Nevertheless, in this study MIS effects are not significantly different between the four groups. On the other hand, management levels could explain a part of the variation in MIS effects between farms. Farmers with high management levels, which as a starting point tend to get better informed than farmer with low management levels, get more added value of MIS.

8. Discussion

The scarcity of studies reporting a numerical or an economical value of information provided by monitoring systems, DSS and other MIS can be discussed. Half of the studies reporting an economical value are from the early stage of the development of monitoring systems and DSS (late 1980s, early 1990s). When an economical value was reported, the units in which this value is expressed make the results difficult to compare. This has not been done in this study, in which scope was limited to review the information and associated values. Nevertheless, it should be noted that with time, the management and monitoring systems have developed and the production efficiency increased, which would make the outcome of these studies (covering more than 20 years) difficult to compare. Later on, as the technology evolved, the development of novel IT systems increased. Nevertheless, the reported expected benefits are the following: (i) mostly described as general matters and necessity (increase productivity for

facing competitiveness) and (ii) to develop systems to cope with intensive production environment (monitor production diseases and welfare).

The lack of reported numerical and economical value may be due to the fact that this value is difficult to estimate because of differences in existing management systems or of the lack of tools to estimate it. Further perspectives could include the development of such a tool, which could, for instance, be a model frame with a detailed level of information that allows one to estimate the value of including or excluding one piece of information. The alternative would be to do it empirically, with the drawback of being to a much higher expense. Another potential explanation for the lack of reported economical values may be that this value is in fact, in most cases, rather low. In that sense, there is little interest in calculating it as such, nor in reporting it, either from the researcher perspective or from a company developing a new system. A potential rather low value of information could be explained by the already achieved progresses in terms of technology and management. These have enhanced the production efficiency over the years and hence, likely narrowed the potential for improvement, in terms of economical value, or benefits. Nevertheless, in a situation with increasing herd sizes and growing competition from countries with lower production costs, the profitability of pig herds will become increasingly sensitive to even small fluctuations in production results. Therefore, even small marginal improvements achieved through, for instance, the use of monitoring systems and DSS, can have an important impact on the productivity.

According to the authors, reporting the numerical value, in terms of e.g. number of piglets, daily gain or labor hours, might be more informative than reporting an economical gain. Firstly, because it allows for comparing the value of information over time, across production systems and across countries, where, for instance, labor cost is very different. Secondly, expressing a gain in terms of the number of piglets or hours per week may be easier to relate to. The preference of reporting numerical values is also supported by the fact that if more focus is put on monitoring traits with higher economical weights, the values of these traits might change in the future. According to [Houska et al. \(2004\)](#), expecting an increase in costs, reproduction and growth traits should be given higher economic traits. On the other hand, expecting a decrease in meat price, production traits will be more important.

It should also be mentioned that more general benefits (in terms of production, welfare or labor) of monitoring systems, DSS and other MIS should not be underestimated. [van der Stuyft et al. \(1991\)](#) mentioned that the developed systems should have a significant effect on production or welfare. Later on, addressing the ethical issues raised by the development of automation systems, [Cornou \(2009\)](#) also emphasized that animal welfare should be addressed while developing these systems, if they should be sustainable. Therefore, even though some systems do not have a directly measurable economical impact, attention should be paid to their other benefits, of which some of them may indirectly impact the productivity, e.g. by prevention or early identification of diseases.

For practical implementation of systems providing new information, [Sonka \(1985\)](#) early suggested that the use of information requires flexibility in management, described as a need for a '*continual reevaluation of management plans as the producer's decision environment changes*'. The influence of the human factor on the use, and then, to a certain extent, value of information (see [Fig. 1](#)) is rarely mentioned for pig production. This aspect has been more discussed for other agricultural production, where information about, for instance, climate forecasts ([Byerlee and Anderson, 1982](#); [Llewellyn, 2007](#); [Mazzocco et al., 1992](#)), have been used for a longer period of time. [Llewellyn \(2007\)](#) describes the adoption decision-making process for agronomic innovations in Australia. Two limited on-farm resources, namely time and the capacity to integrate new information, are described as hindering the decision of adopting innovations. However, the author emphasizes that readily available information with high reliability and relevance to the decision-maker can potentially reduce information and learning costs.

The use and value of the information provided by these systems is ultimately in the user's hand. Focus should therefore be in making the relevant information available. This includes providing the most important key figures only so that the user does not drown in information. It also implies an easy display where the information can be rapidly accessed. Moreover, priority should be given to real time information, or weekly update of results, according to the type of information. For instance, water and feed consumption or temperature should be monitored on a minute or hourly basis, while farrowing rate results could be presented on a weekly basis. To get objective results in real time may be a motivating factor for the manager and employees, and allow them to rapidly take the necessary actions, enhancing thereof the value of the information. To obtain frequent objective results can also support discussions with farm consultants and veterinarians. Therefore, it can be suggested that the development of new monitoring systems, DSS and other MIS should focus on finding out which figures are the most valuable and important, and on developing tools providing information in real time.

9. Conclusion

In conclusion, the findings of this paper indicate that the value of information mostly covers general aspects, such as improving production and management, animal health and welfare, and reduction of labor time and cost. Reporting the value of information in terms of numerical value, such as improvement in daily gain or reduction of working hours, may be more appropriate than reporting an economical value. To quantify the value of information, it is suggested to develop a model frame containing a detailed level of information, where the value of including or excluding one piece of information could be estimated. Development of new monitoring systems, DSS and other MIS should take into account the end user, i.e. the farmer. The information provided should therefore be relevant and in real time, if it should be used for improving productivity, health and welfare.

Conflict of interest statement

The authors report that there is no conflict of interest relevant to this publication.

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